Preview: set-01/by_level/Grade_03-04_Ecolier/y2024_q14 (v3)

Counting chains systematically: list what fits after what, then branch like a tree

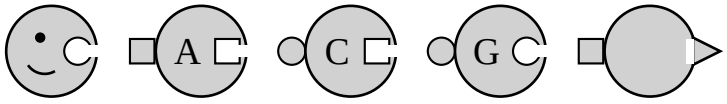
How to think about it: counting all the ways, without missing one

In these questions you build a chain (a caterpillar, a train, ...) from pieces that must fit together, and you have to find out **how many** different chains there are. If you just try pieces in any order, it is very easy to forget one or to count one twice.

The big idea: don't guess. First find out **what can come right after each piece**. Then build every chain from the start, **one branch at a time**, like a tree.

Step by step: the contest question

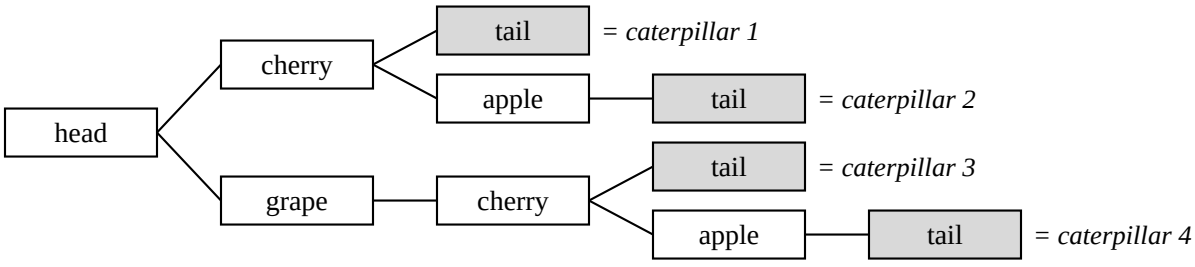
Lucas has a head, a tail and three middle pieces: apple (A), cherry (C) and grape (G). A peg only fits into a hole of the same shape.



- **1. Make a list: what fits right after each piece?** Look at the hole on the right of the piece, then find all pegs of the same shape.

Piece	Its hole	What can come next
head	round hole	cherry, grape
apple	square hole	tail
cherry	square hole	apple, tail
grape	round hole	cherry

- **2. Grow a tree from the head.** Write every piece that can come next as a new branch. Then do the same for each branch, until you reach the tail.



- **3. Count the branches that end with the tail.** Here there are **4** caterpillars.

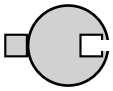
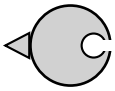
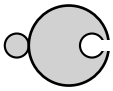
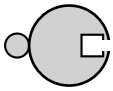
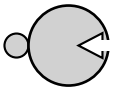
Watch out

- Every piece can be used **only once** in a chain. Cross out pieces you already used on a branch.
- A branch that gets stuck before the tail (nothing fits) is a **dead end**: don't count it.
- Count only **finished** chains, with the head at the start and the tail at the end.

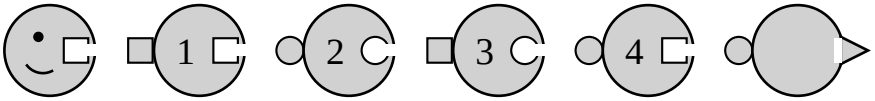
Try it yourself: cut out paper pieces and draw a different peg on each side. Make a tree on paper first, then lay all the chains you found. Did the tree find every one?

Warm-up: small steps first (no points)

W1. A peg fits only into a hole of the **same shape**. Which piece can go right after the head?

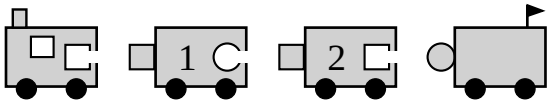
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

W2. A peg fits only into a hole of the same shape. How many of the other pieces (also the tail) could go right after part 4?



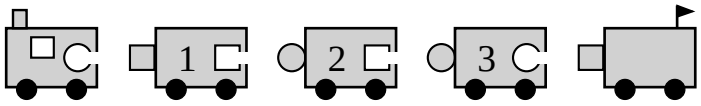
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

W3. Lucas has these pieces: an engine, a last wagon and 2 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1 or 2 numbered wagons, then the last wagon. Each piece can be used only once. How many different trains can Lucas build?



A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

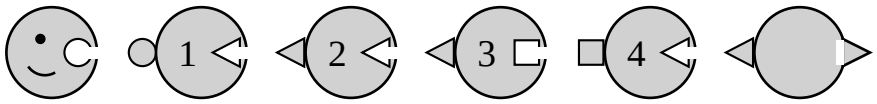
W4. Tom has these pieces: an engine, a last wagon and 3 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2 or 3 numbered wagons, then the last wagon. Each piece can be used only once. Tom puts wagon 3 right after the engine. How many ways are there to finish?



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

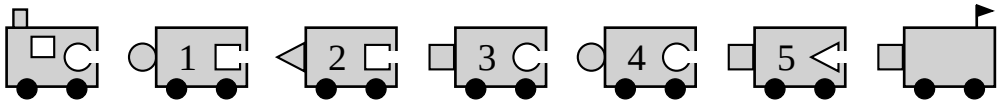
Part A: Each question is worth 3 points

1. Tom has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 numbered parts, then the tail. Each piece can be used only once. How many different caterpillars can Tom build?



A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 8 ***E.*** 10

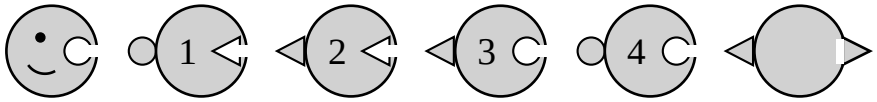
2. Leo has these pieces: an engine, a last wagon and 5 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3 or 4 numbered wagons, then the last wagon. Each piece can be used only once. How many different trains can Leo build?



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 6

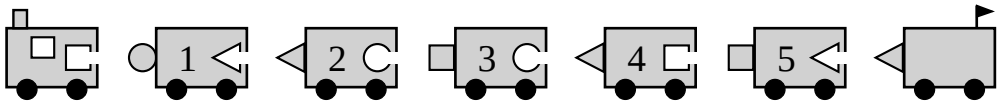
Part B: Each question is worth 4 points

3. Leo has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3 or 4 numbered parts, then the tail. Each piece can be used only once. How many different caterpillars can Leo build with **exactly 2** numbered parts **between** the head and the tail?



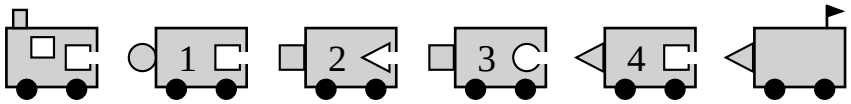
A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

4. Emma has these pieces: an engine, a last wagon and 5 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3, 4 or 5 numbered wagons, then the last wagon. Each piece can be used only once. How many different trains can Emma build with **exactly 4** numbered wagons **between** the engine and the last wagon?



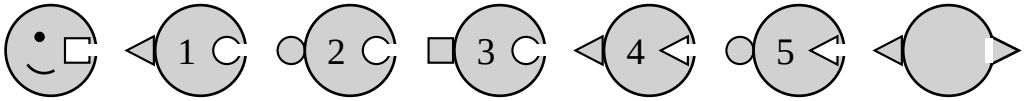
A. 0 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

5. Anna has these pieces: an engine, a last wagon and 4 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3 or 4 numbered wagons, then the last wagon. Each piece can be used only once. Which wagon can **never** be part of one of Anna's trains?



A. wagon 1 ***B.*** wagon 2 ***C.*** wagon 3
D. wagon 4 ***E.*** each wagon can be used

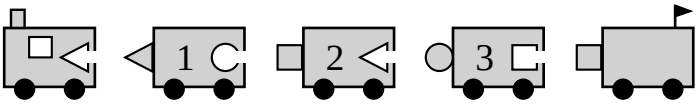
6. Tom has these pieces: a head, a tail and 5 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3, 4 or 5 numbered parts, then the tail. Each piece can be used only once. Which part can **never** be part of one of Tom's caterpillars?

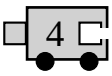


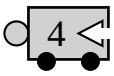
A. part 1 ***B.*** part 2 ***C.*** part 3 ***D.*** part 4 ***E.*** part 5

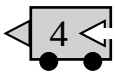
Part C: Each question is worth 5 points

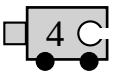
7. Lucas has these pieces: an engine, a last wagon and 3 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3 or 4 numbered wagons, then the last wagon. Each piece can be used only once. Now Lucas can build 1 train. Lucas gets one more wagon (number 4) and then can build exactly 4. Which wagon did Lucas get?

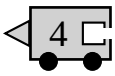


A. 

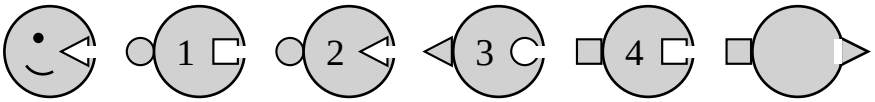
B. 

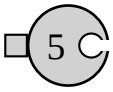
C. 

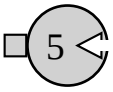
D. 

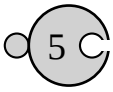
E. 

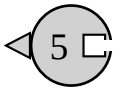
8. Leo has these pieces: a head, a tail and 4 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3, 4 or 5 numbered parts, then the tail. Each piece can be used only once. Now Leo can build 2 caterpillars. Leo gets one more part (number 5) and then can build exactly 6. Which part did Leo get?

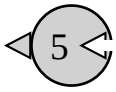


A. 

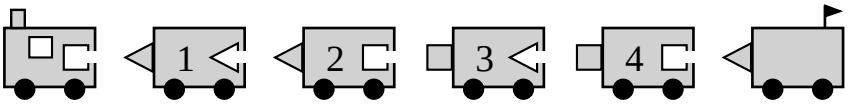
B. 

C. 

D. 

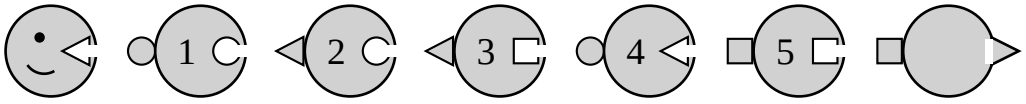
E. 

9. Tom has these pieces: an engine, a last wagon and 4 wagons with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A train is the engine, then 1, 2, 3 or 4 numbered wagons, then the last wagon. Each piece can be used only once. Tom has already put wagon 4 right after the engine. In how many different ways can Tom finish the train?



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

10. Anna has these pieces: a head, a tail and 5 parts with numbers. A peg fits only into a hole of the **same shape**, and the pieces cannot be turned around. A caterpillar is the head, then 1, 2, 3, 4 or 5 numbered parts, then the tail. Each piece can be used only once. Anna has already put part 2 right after the head. In how many different ways can Anna finish the caterpillar?



A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

Answer Key

W1.	**A**	**W2.**	**C**	**W3.**	**C**	**W4.**	**B**	**1.**	**A**	**2.**	**D**
3.	**C**	**4.**	**B**	**5.**	**D**	**6.**	**A**	**7.**	**E**	**8.**	**D**
9.	**B**	**10.**	**B**								

Solutions

W1. A — The head has a square hole, so the next piece needs a square peg on its left: A. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · w_fits_after]

W2. C (2) — Part 4 has a square hole. Count the pieces with a square peg: 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · w_how_many_next]

W3. C (2) — Start at the engine and try every piece that fits, one branch at a time (H = engine, T = last wagon): H–1–T, H–2–1–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · w_small]

W4. B (2) — Follow only this branch: H–3–2–T, H–3–2–1–T. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · w_tree_branch]

1. A (5) — Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–1–T, H–1–2–T, H–1–2–3–4–T, H–1–3–4–T, H–1–3–4–2–T. That is 5. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · count_all]

2. D (4) — Start at the engine and try every piece that fits, one branch at a time (H = engine, T = last wagon): H–1–T, H–1–5–2–T, H–4–1–T, H–4–1–5–2–T. That is 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · count_all]

3. C (2) — Count only the numbered parts, not the head and the tail. Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–1–2–T, H–4–1–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · exact_length]

4. B (2) — Count only the numbered wagons, not the engine and the last wagon. Start at the engine and try every piece that fits, one branch at a time (H = engine, T = last wagon): H–3–1–4–5–T, H–5–4–3–1–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · exact_length]

5. D (wagon 4) — Start at the engine and try every piece that fits, one branch at a time (H = engine, T = last wagon): H–2–T, H–3–1–2–T. That is 2. Wagon 4 is in none of them: it is a dead end, or nothing leads to it. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · never_used]

6. A (part 1) — Start at the head and try every piece that fits, one branch at a time (H = head, T = tail): H–3–2–5–T, H–3–2–5–4–T, H–3–5–T, H–3–5–4–T. That is 4. Part 1 is in none of them: it is a dead end, or nothing leads to it. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · never_used]

7. E — Try each new piece and count again with a tree. With E: H–1–3–T, H–1–3–2–4–T, H–4–T, H–4–2–1–3–T, that is 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · extra_piece]

8. D — Try each new piece and count again with a tree. With D: H–3–1–T, H–3–1–4–T, H–3–2–5–T, H–3–2–5–4–T, H–5–T, H–5–4–T, that is 6. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · extra_piece]

9. B (2) — Look only at the branch that starts with wagon 4: H–4–3–T, H–4–3–1–T. That is 2. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · start_given]

10. B (4) — Look only at the branch that starts with part 2: H–2–1–4–3–T, H–2–1–4–3–5–T, H–2–4–3–T, H–2–4–3–5–T. That is 4. [set-01/by_level/Grade_03-04_Ecolier/2024 Q14 · start_given]